

Harshit Bhavin Kothari

Software Engineer • IIT Gandhinagar Alumnus

+91 7019123702

LinkedIn

Portfolio

harshitbhavin@gmail.com

PROFESSIONAL SUMMARY

- **Software Engineer** and **IIT Gandhinagar** alumnus with experience in building scalable software systems, **AI/ML solutions**, **Computer Vision applications**, and secure automotive platforms.
- Experienced in designing and developing **AI-driven applications**, including **NLP**, **Retrieval-Augmented Generation (RAG)**, agentic AI systems, and machine learning models for real-world business use cases.
- Strong expertise in **CA security**, **PKI infrastructure**, **mTLS**, **CRL**, **RBAC**, **MFA**, and cloud-based security solutions using **AWS Private CA**.
- Skilled across the complete software development lifecycle, including architecture design, API development, frontend engineering, backend systems, automation, and deployment using modern cloud technologies.

EDUCATION & CERTIFICATIONS

Qualification	Specialization / Certification	Institute	Year	Score
B.Tech.	Electrical Engineering	IIT Gandhinagar	2021–2025*	8.08
Class XII	Physics, Chemistry, Mathematics	Shri Chaitanya Techno School, Bangalore	2020–2021	96.8%
Class X	General	Ryan International School, Bangalore	2018–2019	97.6%
Certificate	Data Analytics	Coding Ninjas	2023	98.5%
Certificate	Data Structures & Algorithms	Coding Ninjas	2023	92.5%
Certificate	Data Science & Machine Learning	Coding Ninjas	2023	94%

*Awarded early graduation in December 2024

TECHNICAL SKILLS

- **Languages:** Python, C++, TypeScript, JavaScript, SQL, Java, QNX, Verilog, MATLAB, Mermaid
- **Frameworks & Libraries:** React, Node.js (Express.js), LangGraph, Pydantic, Tailwind CSS, Tkinter
- **Tools & Cloud Platforms:** Docker, Google Cloud Platform (Cloud Run), Supabase, MongoDB, Tableau, PowerBI, Postman, AWS

PROFESSIONAL EXPERIENCE

- **Software Engineer, GlobalLogic (A Hitachi Group Company)** [July 2025 – Present]
 - Contributed across the complete SDLC of an automotive **OTA cybersecurity platform**, including requirements analysis, system architecture design, Low-Level Design (LLD) documentation, creation of Mermaid sequence diagrams, screen flow design, API specification, React-based portal development, backend implementation, and integration of PKI security mechanisms such as mTLS, CRL, RBAC, MFA, and AWS Private CA services.
 - Developed a **Computer Vision-based automated testing system** for infotainment screens, including black screen detection, audio validation, and KPI analysis; built an interactive live-analysis UI using synchronization and multithreading.
 - Designed a **Python Tkinter-based UI** to automate backend integration between HILS and Codebeamer, improving automotive testing workflows with features like multi-language support and keyboard shortcuts.
- **AI-ML Engineer, Acuver Consulting** [Jan 2025 – July 2025]
 - Implemented **Retrieval-Augmented Generation (RAG) models** to enhance chatbot responses and developed AI-driven dashboards integrating embedding search, NLP, prompt engineering, and analytics.
 - Built time series **forecasting models** (ARIMA, SARIMA, SARIMAX, Prophet) to predict product sales, enabling data-driven order planning and inventory optimization; additionally optimized supply chain routes to improve fulfillment efficiency.
 - Developed a **Retail AI Advisor** using **LangGraph-based agentic AI architecture**, where multiple agents processed user prompts and synthesized customer data, location insights, and weather patterns to recommend optimal products.
- **Trading Analyst, Axxela Research and Analytics** [May – July 2024]
 - Analyzed and **traded futures contracts** in the **EURIBOR** market, gaining proficiency in financial markets, trading mechanisms, and technical indicators like ATR and moving averages.
 - Developed expertise in **risk management strategies**, including hedging techniques to mitigate potential losses and optimize trading decisions.

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PROJECTS

- **Retail Hiring Website | Full Stack Developer** [Dec 2024]
 - Engineered a high-performance, responsive **Full-Stack Talent Hiring Platform** using React (TypeScript, Tailwind CSS) for the frontend, Node.js (Express) & Python for the backend, and MongoDB & Supabase for data persistence.
 - Built an automated assessment engine integrating the **Google Gemini API** via a Python bridge; utilized **Pydantic structured output schemas** to dynamically compile custom multiple-choice question tests from uploaded PDFs, PPTs, or JDs.
 - Developed a daily-active **Gamified Rewards Hub** featuring canvas-confetti and mini-games (Wordle, Hangman, Memory Match) to boost candidate engagement, alongside an India-centric proximity filtration system using a hierarchical pin-code database.
 - Developed a **Progressive Web App (PWA)** with responsive offline-ready capabilities and implemented **real-time notifications** to provide instant updates and improve candidate engagement.
 - Containerized application workloads via **multi-stage Docker builds** and deployed the service to **Google Cloud Run (GCP)**, implementing a secure authentication flow using Google OAuth, JWT, and Nodemailer SMTP.
- **Punjabi Language NLP Model | NLP Research Project, IIT Gandhinagar** [August – November 2024]
 - Collected, cleaned, and curated a large-scale Punjabi language dataset for developing NLP models focused on regional language understanding and generation tasks.
 - Trained and evaluated transformer-based **Natural Language Processing (NLP)** models for Punjabi text understanding, achieving significant results in **sentiment analysis, text classification, and text generation**.
 - Implemented data preprocessing pipelines, model fine-tuning strategies, and performance evaluation techniques to improve Punjabi language model accuracy and contextual understanding.
- **Retail Strategy Analysis | Data Analyst Project, Coding Ninjas** [June 2023]
 - Analyzed multi-table retail datasets (orders, customers, returns) using **SQL** to derive key insights on profitability, state-wise performance, and customer behavior, including return pattern analysis.
 - Built **Tableau dashboards** (pie, line, and map charts) to visualize sales trends and geographic profit distribution, enabling data-driven business decision-making.
- **Data Visionaries, Bitcoin Blockchain analysis, IIT Gandhinagar** [January – April 2024]
 - Conducted graph and nodal analysis on Bitcoin blockchain data using **data science algorithms** like Louvain Algorithm and spectral clustering to uncover patterns, network centrality, and vulnerabilities within transaction flows.
- **LPG Gas Detection and Prevention System, IIT Gandhinagar** [January – April 2024]
 - Developed an LPG leakage and safety alerting platform utilizing an **STM32 microcontroller** and NodeMCU, implementing detection algorithms to actuate servo-valves, trigger buzzers, and push real-time notifications to users.
- **Simulation Study of Solar Cells, IIT Gandhinagar** [August – November 2023]
 - Conducted in-depth structure simulations of solar cells using **nextNano software**, implementing structural optimization techniques that successfully yielded enhanced operational efficiency.

POSITIONS OF RESPONSIBILITY

- **Tech Expo Executive, Amalthea [Annual Technical Summit]** [2022-2023]
 - Managed databases and established communications with corporate HoDs, collaborating effectively within a 10-member team to drive technological presentation layouts and logistics.
- **Event Executive, Blithchron [Annual Cultural Fest]** [2022-2023]
 - Successfully led a team of 10 individuals as the event leader for a singing event – Euphony, showcasing strong management, cross-functional coordination, and event execution logistics.

RANKS AND ACHIEVEMENTS

- Achieved **99.689 percentile in JEE Mains 2021** with an All India Rank of 3515 and a perfect **100 percentile in Mathematics**.
- Secured **All India Rank 663 in GATE 2025 – Data Science & AI** and **AIR 1130 in Electronics & Communication Engineering**.
- Achieved a **96.95 percentile in the CAT exam**, with an exceptional **99.76 percentile in Data Interpretation & Logical Reasoning**.
- Advanced to the **semi-final round of the Tata Imagination Challenge**, a nationwide competition, demonstrating excellence in business problem-solving and technology innovation.
- Won a **bronze medal** at the state level in the **National Cyber Olympiad (NCO)**.

EXTRA-CURRICULAR ACTIVITIES

- Represented IIT-Gandhinagar in the dance competition at **Inter-IIT Cultural Meet-2022** held at IIT Chennai.
- A passionate **sports enthusiast** with a keen interest in cricket, actively participating in the **GlobalLogic Premier League (GPL)** and embracing teamwork, collaboration, and healthy competition. Also enthusiastic about traveling and exploring new places and cultures to broaden perspectives.